An Innovative Framework for Hourly Satellite Soil Moisture Retrieval via Integrated Spatiotemporal Downscaling Techniques

Peilin Song, Mengran Wang, Lixin Dong, Tianjie Zhao, *Senior Member, IEEE*, Haigen Zhao, Jingfeng Huang, Panpan Yao, Jingyao Zheng, and Yongqiang Zhang

***Abstract*—Satellite microwave remote sensing is acknowledged as the primary method for obtaining global-scale surface soil moisture (SSM) data. Typically, spaceborne microwave sensors aboard polar-orbiting satellites yield SSM estimates with a native spatial resolution of several tens of kilometers and a temporal resolution of about once or twice daily. This indicates considerable potential for enhancing both of their spatial and temporal resolutions. Although spatial downscaling of spaceborne microwave SSM has garnered significant attention recently, the enhancement of their temporal resolution has received less focus. This study pioneers a methodology for generating hourly scale satellite SSM estimates. The developed approach integrates a novel blending module that combines geostationary satellite observations with a spatially downscaled SSM dataset derived from traditional fusion among microwave and optical observations on polar-orbiting platforms. This blending module leverages land surface temperature (LST) data from geostationary satellites, which effectively quantify SSM variations on an hourly interval upon the thermal inertia theory. Consequently, a comprehensive spatiotemporal integrated framework for SSM downscaling is established to produce hourly scale SSM at a resolution of 6 km, enhancing upon the daily and 36-km resolutions of existing microwave SSM datasets. Validation of the downscaled hourly SSM estimates was conducted through an established ground soil moisture observatory network in North China, revealing an unbiased root-mean-square error (ubRMSE) of no higher than 0.04 cm^3/cm^3. This result confirms preservation of the fundamental accuracy of original microwave SSM retrievals and demonstrates the effectiveness of the developed framework in improving both spatial and temporal representativeness of SSM data.**

***Index Terms*—Diurnal temperature cycle (DTC), downscaling, geostationary satellite, hourly soil moisture, microwave.**

Received 27 October 2024; revised 28 April 2025, 21 July 2025, 25 August 2025, and 30 October 2025; accepted 18 November 2025. Date of publication 21 November 2025; date of current version 1 December 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFD2300300, in part by the National Natural Science Foundation of China under Grant 42371330, and in part by the Fengyun Satellite Application Pioneering Plan under Grant FY-APP-2024.0401. *(Corresponding authors: Peilin Song; Haigen Zhao.)*

Peilin Song and Mengran Wang are with the Key Laboratory of Physical Electronics and Devices, Ministry of Education, Faculty of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: foreverspl@xjtu.edu.cn; iamwmr@stu.xjtu.edu.cn).

Lixin Dong is with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center, China Meteorological Administration, Beijing 100081, China (e-mail: dlx_water@163.com).

Tianjie Zhao and Jingyao Zheng are with the State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: zhaotj@aircas.ac.cn; zhengjingyao94@gmail.com).

Haigen Zhao is with the Institute of Environment and Sustainable Development in Agriculture, Chinese Academy of Agricultural Sciences, Beijing 100081, China (e-mail: zhaohaigen@caas.cn).

Jingfeng Huang is with the Institute of Applied Remote Sensing and Information Technology, Zhejiang University, Hangzhou 310058, China (e-mail: hjf@zju.edu.cn).

Panpan Yao is with the College of Surveying and Geo-Informatics, North China University of Water Resources and Electric Power, Zhengzhou 450046, China (e-mail: yaopanpan@ncwu.edu.cn).

Yongqiang Zhang is with the Key Laboratory of Water Cycle and Related Land Surface Processes, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China (e-mail: yongqiang.zhang2014@gmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3635555

I. Introduction

SURFACE soil moisture (SSM) plays a crucial role in maintaining sustainable development of Earth civilization as it has significant impacts on different components of the human ecosystem, like agriculture production, climate change, water resource management, weather/climate prediction, as well as hydrological modeling. The temporal variations in SSM are particularly responsive to the conditions and evolving trends of environmental factors within the ecosystem. For instance, the accurate simulation of hydrological models, such as those predicting rainfall-runoff, relies on SSM time-series data for initializing models at the onset of rainstorms and for ongoing data assimilation throughout these events [1], [2], [3], [4], [5]. In addition, due to its effect on surface energy balance, frequent SSM variability data (e.g., at an intradaily scale) can improve climate models, enabling more accurate predictions of short-term weather events like heatwaves or heavy showers [6], [7]. Furthermore, in agricultural practices such as precision farming [8], understanding the temporal fluctuations in SSM is crucial for farmers' real-time decisions on irrigation scheduling [9], [10]. This knowledge helps to optimize water use, preventing both over- and under-watering, and thus enhances crop yields. Moreover, high-frequency monitoring of SSM time-series data can serve as an early warning system for natural disasters such as flash floods or landslides in mountainous regions by detecting unusually rapid changes in SSM levels that often precede these events [11], [12].

Satellite remote sensing has emerged as a crucial non-invasive technique for monitoring SSM across extensive areas, particularly at continental and global scales—something unachievable through ground-based measurement alone. The ability of satellite sensors to penetrate clouds and vegetation

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

canopies makes them indispensable, especially when operating in the microwave frequency range of 1.4–10.7 GHz [13], [14], [15]. This range is currently the primary data source for large-scale, all-weather SSM estimation. Microwave remote sensing is categorized into two main types: active and passive. Passive microwave sensors, such as radiometers, are integral to missions like the Advanced Microwave Scanning Radiometer 2 (AMSR2) [16], the Soil Moisture Active and Passive (SMAP) [17], and the Soil Moisture and Ocean Salinity (SMOS) [18]. On the other hand, active microwave sensors typically include scatterometers, which are particularly relevant at the global scale. Notable examples are the Advanced Scatterometer (ASCAT) aboard the Meteorological Operational (Metop) satellites [19] and the WindRAD instrument on China's FengYun-3E meteorological satellite [20]. Although spaceborne synthetic aperture radar (SAR) systems, such as those on Sentinel-1 [21] or RADARSAT-2 [22], can generate land surface maps with much higher spatial resolutions, their applications for global monitoring are limited due to the overall irregular and poorer revisit frequencies (>5 days).

Spaceborne radiometers and scatterometers, despite their valuable contributions, still face significant limitations in both spatial and temporal resolutions. Spatially, the native resolution of a single observation across the target waveband range typically falls beyond 20 km. While such data may be beneficial for monitoring large-scale phenomena, it is insufficient for applications requiring finer details, such as farm- or field-level analyses in agriculture and hydrology. Therefore, this coarse resolution presents challenges for addressing localized demands with precision [23], [24]. In the temporal domain, microwave sensors on polar-orbiting satellites generally offer revisit cycles of 1–3 days for ascending and descending passes, respectively, in mid-to-low latitude regions. Consequently, the maximum observational frequency of a single sensor is limited to twice daily (once during the ascending pass and once during the descending pass). Despite its superiority to the existing spaceborne SAR systems, this frequency is still inadequate for SSM monitoring on an "intradaily" basis, which is crucial for most of the applications that depend on detecting short-term variations in SSM. As a result, the ability of current satellite observations to support most of the aforementioned applications (introduced at the beginning of this section) that require SSM of high temporal resolution is significantly constrained.

In recent years, the enhancement of spatial resolution in microwave-derived SSM products has become a focal point within the academic community. The primary approach to this enhancement involves "downscaling," which integrates microwave-derived SSM with high-resolution data from auxiliary satellite sources, such as optical, thermal-infrared (TIR), or SAR sensors [25], [26], [27], [28], [29], [30]. Significant progress has been yielded from this integration, enabling the development of long-term SSM datasets with spatial resolutions finer than 5 km and suitable for global or large regional applications [31], [32], [33]. This advancement marks a pivotal shift from traditional microwave-derived SSM. However, despite these strides in spatial resolution, there remains a notable gap in research addressing the improvement of temporal resolution for microwave-derived SSM [34]. Current efforts have largely overlooked the potential for increasing the frequency of satellite microwave SSM data acquisition from the conventional daily intervals to more granular "intradaily" intervals, which could provide much-needed insights for more dynamic environmental monitoring.

Complementing conventional microwave remote sensing, spaceborne Global Navigation Satellite System Reflectometry (GNSS-R) has recently emerged as a cutting-edge technique for large-scale SSM monitoring. This innovative approach operates in a bistatic radar configuration, utilizing forward-scattered GNSS signals captured by dedicated low-Earth-orbit (LEO) receivers. Notably, GNSS-R surpasses traditional microwave-based methods in temporal sampling capability, offering unprecedented revisit frequencies for SSM estimation [35], [36]. A prime example is the Cyclone Global Navigation Satellite System (CYGNSS) [37], a pioneering GNSS-R mission designed for Earth surface parameter retrieval, which achieves an exceptional intradaily temporal resolution as frequent as 7.2 h locally. However, the current orbital design of CYGNSS results in strong spatial heterogeneity in its global performance, with observation density decreasing significantly as latitudes increase from the equator to the poles [38]. In addition, the development of GNSS-R-based SSM retrieval algorithms relies heavily on low-temporal-resolution passive microwave SSM data or model-simulated soil moisture data [39], [40], [41]. This dependence significantly undermines the inherent advantage of satellite GNSS-R in providing high-temporal-resolution SSM observations. Consequently, their average revisit intervals remain constrained to daily scales at present [37], [42], highlighting the ongoing need for improving the temporal resolution of satellite-derived SSM with greater algorithmic universality across the globe.

Building upon these advancements, this study introduces a novel multisource satellite data fusion framework aimed at enhancing the temporal resolution of microwave-derived daily SSM products to consistent hourly scales. The proposed approach extends our previously developed spatial downscaling methodology into a unified time–space integrated framework. Specifically, it synergistically combines microwave-derived SSM from one polar-orbit platform with TIR observations obtained from another geostationary satellite. By leveraging the high temporal resolution of geostationary sensors alongside the daily SSM retrievals from polar-orbiting microwave missions, we investigate the feasibility of producing temporally continuous, hourly SSM estimates. This strategy provides a practical interim solution ahead of future missions specifically designed for intradaily SSM observations, such as the ESA Geosynchronous-Continental Land Atmosphere Sensing System [43] and the Copernicus Imaging Microwave Radiometer [44]. Furthermore, as one of the pioneering efforts in this direction, this study aims to establish a foundation for subsequent research on intradaily scale temporal analyses of satellite-derived SSM. The anticipated improvements in temporal resolution are expected to significantly advance applications in precision agriculture,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

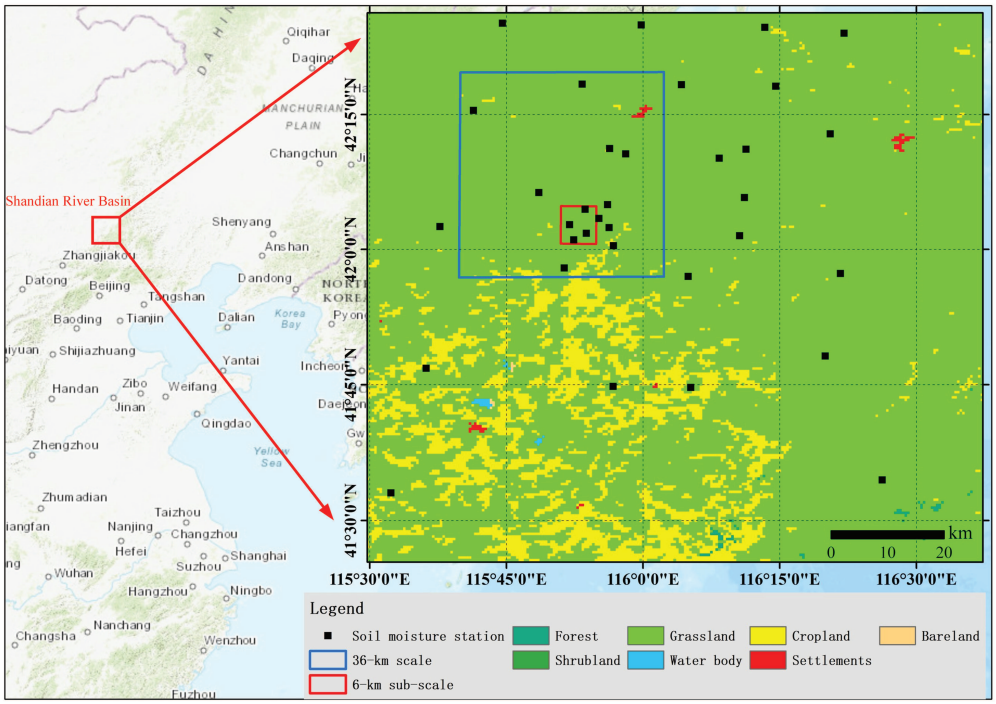


Fig. 1. Spatial distributions of the 34 soil moisture stations in the SMN-SDR. The background land cover is from the MODIS MCD12Q1 land cover map in 2019. The rectangles colored in blue and red denote different sampling scales, respectively, at 36 and 6 km for satellite SSM evaluation (see Section II-B for details).

hydrological and climate modeling, and ecosystem management by delivering more dynamic and temporally refined soil moisture information.

II. Methods and Materials

A. Study Area and Ground Measurements

The study focuses on the Shandian River Watershed, which traverses Hebei and Inner Mongolia in North China (see Fig. 1). This watershed falls within a temperate continental climate zone, characterized by an annual average rainfall ranging between 300 and 500 mm and predominantly concentrated in the summer months. The area's relatively flat topography supports diverse land cover types contributing to its ecological variability, including grasslands, croplands, shrublands, and forests. These land cover patterns play a critical role on influencing the hydrological property of the watershed [45].

The primary rationale for selecting this study area is the presence of an established wireless sensor network for monitoring soil moisture, hereafter referred to as SMN-SDR, which was implemented during the SMELR project between July 18 and September 28, 2018 [46]. As illustrated in Fig. 1, the SMN-SDR comprises 34 stations spreading across approximately 10 000 km^2 (115.5°E–116.5°E and 41.5°N–42.5°N). Each station is equipped with five Decagon 5TM sensors installed horizontally at various depths (3, 5, 10, 20, and 50 cm) to monitor soil moisture and related variables, such as soil temperature, at continuous 10-min intervals since 00:00 UTC on September 8, 2018. Further details regarding the sensors are available in Zhao et al. [46]. In addition, the integration of SMN-SDR into the International Soil Moisture Network (ISMN) [47] enhances its significance as a reliable source of ground-based data for validating various spaceborne SSM products [45].

B. Time–Space Integrated Downscaling Framework and Its Satellite Data Inputs

The proposed time–space integrated framework for SSM downscaling consists of two hierarchically organized submodules (see Fig. 2). The first submodule utilizes optical and TIR data from Aqua MODIS to refine the spatial resolution of microwave SSM, which is observed at quasi-daily intervals. This process, commonly referred to as "spatial downscaling," improves the spatial granularity of the data. While this study does not conduct specific experiments for this submodule, it leverages results from our prior research, which applied advanced spatial downscaling techniques [28], [32] to an AMSR2 SSM dataset originally at a 36-km resolution, formatted in EASE-GRID-2.0. This AMSR2 SSM product (https://doi.org/10.11888/Soil.tpdc.270960) was developed using an artificial neural network model trained against SMAP 36-km SSM data, achieving a comparable global accuracy (approximately 0.05 cm^3/cm^3 in RMSE) to its SMAP counterpart [48]. The spatially downscaled SSM product, generated through our established methodology, exhibits an unbiased root-mean-square error (ubRMSE) ranging between 0.05 and 0.08 cm^3/cm^3. Spanning the period from 2003 to 2023, this product offers a 1-km resolution akin to MODIS projections, with a consistent daily revisit frequency covering all continental regions of China. The dataset is publicly accessible via the National Tibetan Plateau Center (http://doi.org/10.11888/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

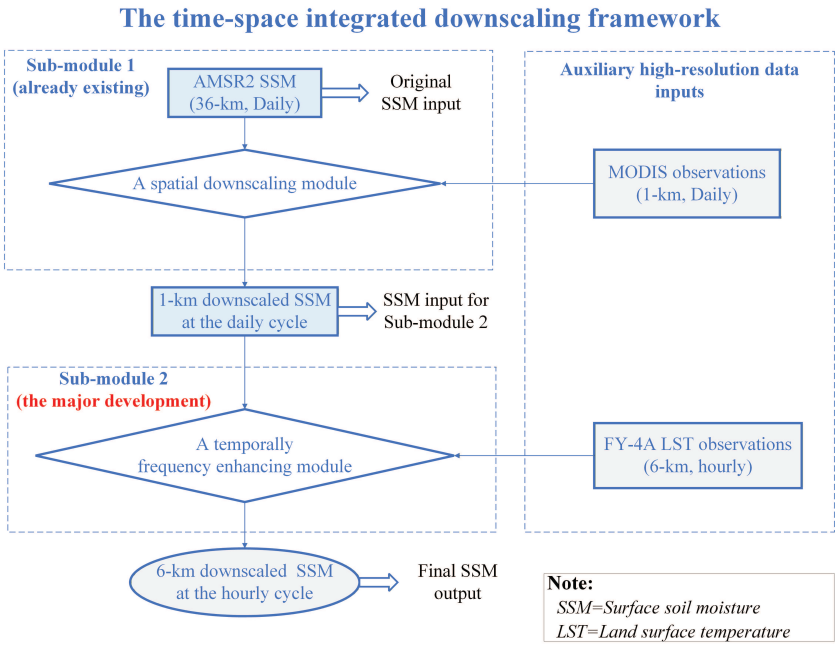


Fig. 2. Flowchart of the proposed time–space integrated downscaling framework, primarily demonstrating the relation among different data inputs/outputs and different submodules.

Hydro.tpdc.271762). This product serves as a foundational input for subsequent analyses in the second submodule, while detailed methodologies of the spatial downscaling process are available in our earlier studies [28], [32].

The second submodule focuses on augmenting the temporal frequency of the spatially enhanced SSM, transforming it into a refined product that surpasses the original microwave (AMSR2) SSM in both spatial and temporal dimensions. The core innovation of the entire framework lies in the development of this submodule, which aims to elevate the temporal resolution of the spatially downscaled daily SSM data to an hourly cycle. Here, the spatially downscaled SSM from the first submodule serves as a foundational input, providing the baseline soil moisture information necessary for temporal refinement. The high temporal resolution required to achieve the target specifications is primarily enabled by the FengYun-4A (FY-4A) land surface temperature (LST) dataset, which supplies dynamic land surface information at an hourly scale. Launched by China and operational from 2016 to 2023, FY-4A is a geostationary meteorological satellite observing the Asia–Pacific region, with its central observation point at 104°E longitude. The LST data are obtained through TIR signals captured by the Advanced Geosynchronous Radiation Imager (AGRI), which is analogous to the MODIS sensor [49]. A previous study by Dong et al. [50] demonstrated the high accuracy of the FY-4A LST dataset, reporting an RMSE between 1.5 and 3 K when compared with both MODIS and ground-based LST measurements. For specific validation of the FY-4A LST in our study area, please refer to the Appendix. As a geostationary satellite, FY-4A provides a native spatial resolution of approximately 4 km at the equator, which gradually degrades to about 6 km toward higher latitudes. To ensure spatial consistency across all regions and support global-scale applications, the original full-disk FY-4A LST data were uniformly resampled to 6-km resolution, so as to match the coarsest native resolution observed at high latitudes.

Notably, in the first submodule, the original AMSR2 daily SSM product is derived from observations at descending orbital nodes, while the Aqua MODIS data, serving as high-resolution ancillary inputs for spatial downscaling, are acquired during its nighttime overpass. Due to their shared local overpass time of approximately 1:30 A.M., the spatially downscaled 1-km SSM product also inherits this stable temporal reference. In contrast, SMAP observations occur at ~6:00 A.M. and P.M., misaligned with the MODIS overpass time. Consequently, directly using SMAP as the coarse-resolution input would result in a downscaled product with temporally inconsistent observation references, thereby undermining the fidelity of subsequent temporal enhancement. Given that a central objective of this study is to investigate subdaily SSM dynamics, preserving a consistent observation time across input datasets is critical. Furthermore, although SMAP retrievals offer nominally higher accuracy, they suffer from frequent data gaps due to a narrower observational swath width [20]. In contrast, the AMSR2-based SSM product derived via a neural network trained against SMAP data to a large degree retains the spatial accuracy of SMAP while providing superior temporal continuity. However, to ensure a fair assessment of the uncertainties associated with this selection, we additionally incorporated the SMAP dual-channel algorithm SSM product (descending node, ~6:00 A.M. local time; DOI: https://dx.doi.org/10.5067/4XXOGX0OOW1S) into the evaluation framework alongside other datasets used in this study.

Finally, by integrating the spatially downscaled SSM with the hourly FY-4A LST inputs in the second submodule, SSM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

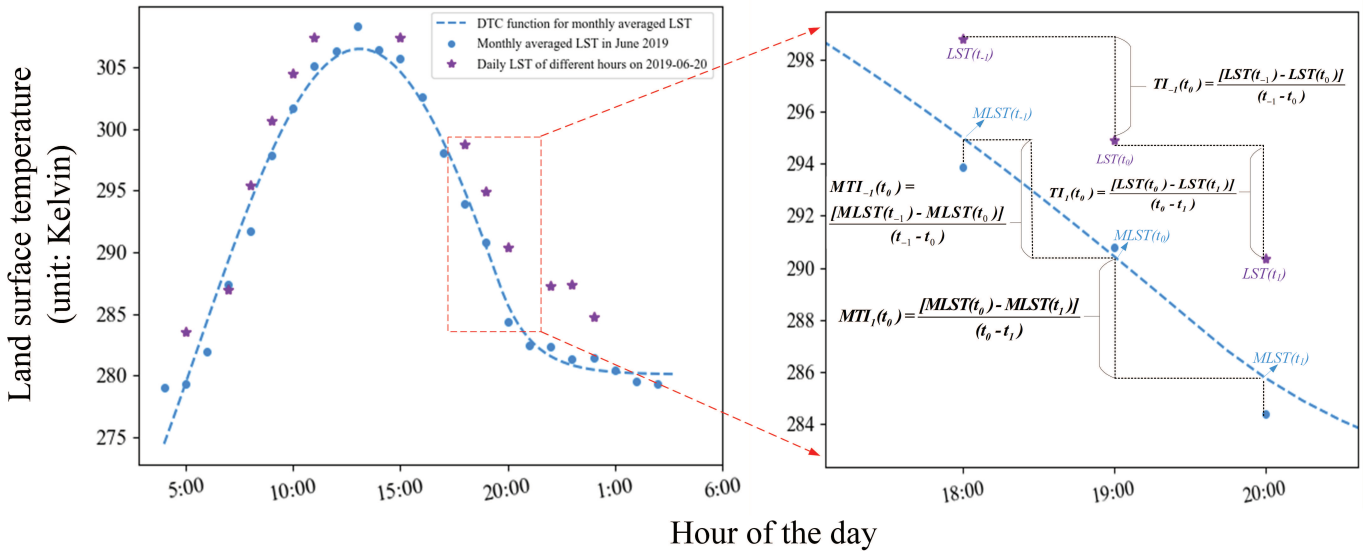


Fig. 3. Illustration of how TI is established at the hourly scale, using FY-4A hourly LST time series on June 20, 2019 as an example.

can be estimated at a 6-km spatial resolution and an hourly temporal scale, significantly improving upon the 36-km resolution and daily revisit frequency of the original AMSR2 SSM. Theoretical and implementation details of this submodule are further elaborated in Sections II-C and II-D.

C. Characterization of the Relative Thermal Inertia Degree of Surface Soil at the Hourly Scale

Enhancing the frequency of daily scale SSM data can be conceptualized as a form of "temporal downscaling." This process parallels traditional spatial downscaling methods, such as the "disaggregation based on physical and theoretical scale change (DISPATCH)" [25], [51], [52] and the "triangle feature space (TFS)" [53]. A crucial aspect of this approach involves establishing a relationship between daily scale SSM and auxiliary datasets that offer high-frequency information, such as the FY-4A LST data. Previous researches [29], [54] indicate that the variation between daytime and nighttime LST can be effectively employed to quantify the thermal inertia properties of the land surface. These properties exhibit a strong sensitivity to variations in soil moisture on a daily scale. Expanding upon this idea, the following paragraphs will explore methods for quantifying the "relative thermal inertia degree" of surface soil at an hourly resolution. This exploration will contribute to deepening our understanding of the intradaily dynamics of surface soil thermal inertia, thereby facilitating the incorporation of thermal data from geostationary satellites into soil moisture modeling systems.

To demonstrate the establishment of the "thermal inertia index" (hereafter referred to as TI) at an hourly scale, we utilized the FY-4A satellite's hourly LST time series, represented by the purple-colored points in Fig. 3. This time-series dataset was collected on June 20, 2019, within the target 6-km grid located in our study area. The calculation of TI at a given time (t_0) relies on a comparison of LST values at three consecutive time points: t_{-1} (the previous hour), t_0 (the current hour), and t_1 (the subsequent hour). TI at t_0, denoted as TI(t_0), is computed as the first-order derivative of the discrete LST time series at that moment. The mathematical formulation of TI(t_0) is presented in the following equations:

$$\mathrm{TI}(t_0) = 0.5^{*}[\mathrm{TI}_{-1}(t_0) + \mathrm{TI}_{1}(t_0)] \tag{1}$$

$$\mathrm{TI}_{-1}(t_0) = [\mathrm{LST}(t_{-1}) - \mathrm{LST}(t_0)]/(t_{-1} - t_0) \tag{2}$$

$$\mathrm{TI}_{1}(t_0) = [\mathrm{LST}(t_1) - \mathrm{LST}(t_0)]/(t_1 - t_0). \tag{3}$$

Theoretically, TI(t_0) provides insight into SSM at the specific moment t_0. However, it is important to recognize that diurnal and seasonal variations in LST, driven by predictable changes in solar radiation, can introduce additional variability into TI. In other words, as t_0 shifts across different times of day or seasons, the responsiveness of TI to SSM may not remain consistent. To address this inconsistency, we drew inspiration from the concepts of soil moisture anomaly [55], [56] and precipitation anomaly [57], which focus solely on the "anomaly component" of the target variable by subtracting its "climatological component" over a relatively long period. Building on this idea, we introduced a "TI anomaly (TIA)," derived from TI, as the difference between TI(t_0) and its monthly average, denoted as MTI(t_0). The formula is expressed as

$$\mathrm{TIA}\,(t_0) = \mathrm{TI}\,(t_0) - \mathrm{MTI}\,(t_0)\,. \tag{4}$$

Here, MTI(t_0) represents the "climatological component" for all TI values at the moment t_0 within a given month. Similar to the rationale behind soil moisture anomaly or precipitation anomaly, TIA is theoretically more comparable across different moments within a diurnal cycle, as it removes the influence of the "climatological component."

To compute MTI(t_0) in (4), the following steps were previously undertaken. First, the monthly average LST values were calculated for each hour of the day (e.g., 0:00, 1:00, . . ., 23:00), as illustrated by the blue points in Fig. 3. In the subsequent step, these data points were fit using a standard "diurnal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

temperature cycle (DTC)-based" function, represented by the blue lines in Fig. 3. Further details on the DTC method can be found in [58] and [59]. Finally, MTI(t_0) was determined in a manner similar to the computation of TI(t_0), following (1)–(3), but using the LST values from the DTC-fit time series rather than the original daily LST observations.

With TIA now calculated at the hourly scale, our subsequent analyses focus on investigating the quantitative relationship between soil moisture and this newly developed indicator. This relationship will serve as the basis for constructing the aforementioned submodule dedicated to temporal downscaling.

D. Implementation of Generating Satellite SSM at the Hourly Scale

Similar to the TI used to quantify variations in SSM, the TIA developed through (4) can also serve as a clear indicator for the monthly anomaly of SSM at any given moment t_0, referred to as SSMA(t_0). This is expressed as

$$\mathrm{SSMA}\,(t_0) = \mathrm{SSM}\,(t_0) - \mathrm{M_SSM}\,(t_0)\,. \tag{5}$$

In line with the definition of MTI(t_0), M_SSM(t_0) is determined by averaging all SSM observations taken at t_0 throughout the month. In Fig. 4(a), ground-based soil moisture measurements aggregated within the study area's 6-km target pixel are shown as an example to explore the monthly variation of M_SSM, with t_0 set between 0:00 and 23:00 for different months in 2019. In addition, Fig. 4(b) illustrates the standard deviation values of M_SSM at different times of the day. Notably, the maximum standard deviation, observed in March, is less than 0.01 cm^3/cm^3 (approximately 0.004 cm^3/cm^3). These findings indicate that M_SSM remains generally stable across various hourly moments, regardless of monthly or seasonal changes.

Building on the insights provided by Fig. 4, the submodule designed to estimate fine-resolution SSM at an hourly scale is outlined through the flowchart in Fig. 5. The specific steps involved in this process are detailed below.

(i) *Spatiotemporal Matching of Satellite Data Inputs:*

This step applies to both FY-4A LST data and the spatially downscaled SSM input at a 1-km resolution. The native 15-min interval of the FY-4A LST product must be aggregated to align with a standard hourly scale. For instance, the LST value at any given hour (denoted as *XX*:00, where "*XX*" is any integer from 1 to 23) is derived by averaging the FY-4A LST measurements taken at five time points: (*XX*−1):30, (*XX*−1):45, *XX*:00, *XX*:15, and *XX*:30. It is important to note that LST values affected by cloud contamination are excluded from this averaging process. Concurrently, the 1-km spatially downscaled SSM data is upscaled to a 6-km resolution by averaging the SSM values of all 1-km subpixels, to ensure spatial consistency with the FY-4A LST product.

(ii) *Modeling SSMA Based on TIA:*

This step focuses on the 6-km target pixel within the study area, where SSMA is computed using spatially aggregated ground-based soil moisture measurements. These values are correlated with TIA derived from FY-4A LST. Given that FY-4A LST can be affected by cloudy conditions, TIA is only

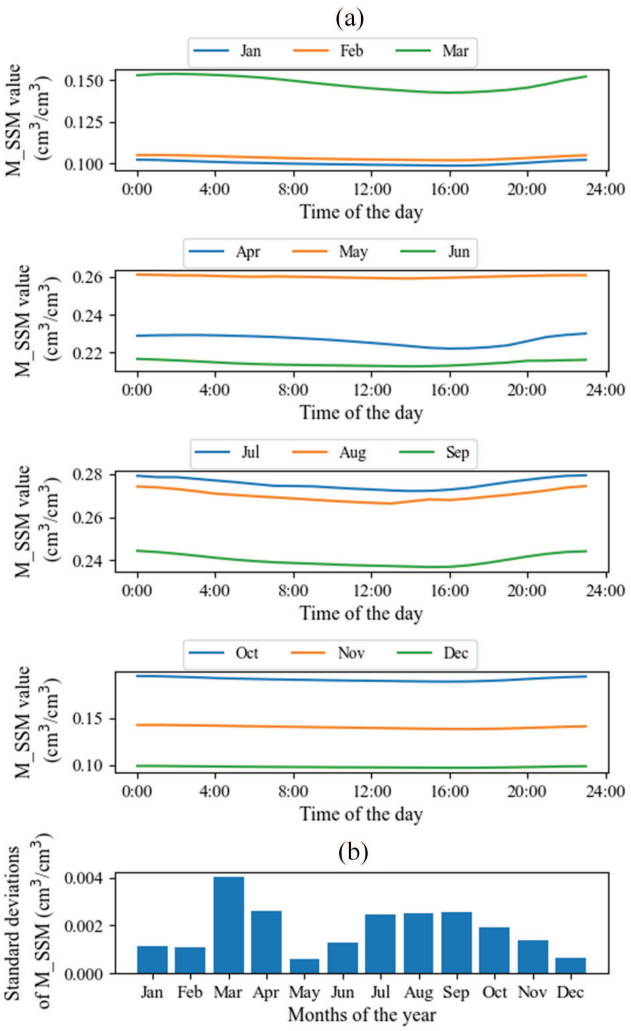


Fig. 4. (a) Variation of M_SSM with respect to different hours between 0:00 and 23:00 for different months in the year 2019. (b) Standard deviation values of M_SSM with respect to different hours, for different months in the year of 2019.

calculated for days when there are more than ten valid (clear sky) FY-4A LST estimates (with a maximum of 24 estimates) per day. According to Fang et al. [60], soil moisture variation can be locally modeled as a linear function of land surface thermal inertia, particularly over short periods (such as within a month). Therefore, a linear relationship between SSMA and TIA at any given time t_0 is assumed in this study, expressed as

$$\mathrm{SSMA}\,(t_0) = a \times \mathrm{TIA}\,(t_0) + b. \tag{6}$$

Here, a and b are coefficients that are fit for each month. The equation's coefficients of determination (R^2) are calculated to assess its feasibility for each month.

(iii) *Estimating Satellite-Based SSMA:*

Using the 1-km spatially downscaled SSM input (which has been aggregated to a 6-km resolution), SSMA at 1:30 A.M. local time is estimated to match TIA, which is also derived from FY-4A LST at 1:30 A.M. This allows the coefficients in (6) to be calculated for each pixel and each month. The next step involves calculating TIA(t_0) from FY-4A LST for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

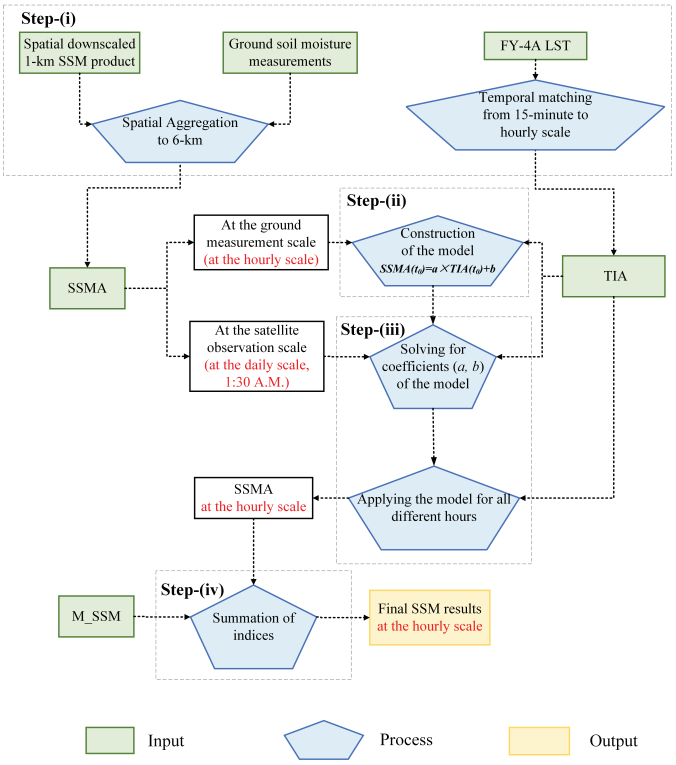


Fig. 5. Detailed flowchart of the "submodule 2 (as highlighted in Fig. 2)" for enhancing the temporal frequency of the spatially downscaled SSM.

any integer hour (from 0:00 to 23:00) through iteration, which serves as the input for (6). This method provides satellite-based estimates of SSMA at an hourly scale for any 6-km pixel.

(iv) *Retrieval of SSM at the Hourly Scale:*

The 6-km SSM for any hour t_0 is retrieved by adding SSMA(t_0) to M_SSM(t_0). Based on the analysis from Fig. 4, M_SSM is consistent across different hourly moments and is therefore considered constant for all hours. It is determined by averaging the spatially downscaled SSM inputs (at 1:30 A.M. local time) for all clear-sky days in the month. Notably, SSM inputs affected by cloudy conditions are excluded from the calculation of M_SSM, ensuring that the index remains consistent with MTI in (4), which is based solely on clear-sky LST data.

E. Validation Strategy on the Developed Framework

Our study validated the hourly scale SSM product by comparing it with ground-based soil moisture measurements, as outlined in Section II-A. To align the spatial scales of satellite and ground data, we aggregated top-layer (3 cm) soil moisture measurements from four ground stations located within the same 6-km pixel, as illustrated by the red grid in Fig. 1. The larger 36-km pixel encompassing this 6-km grid, corresponding to the AMSR2 product, is highlighted in blue in Fig. 1. The experimental dataset, including satellite observations and ground measurements, spans the entire year of 2019.

To ensure a comprehensive evaluation of our proposed spatiotemporal downscaling framework, we included both the original AMSR2 SSM product at 36-km resolution and the downscaled 1-km SSM product (output of submodule 1) for supplementary validation. The downscaled 1-km SSM product was resampled to 6-km resolution and validated within the red grid apart from the finally derived hourly scale SSM. For the AMSR2 SSM evaluation, soil moisture data from 14 ground stations distributed across the 36-km pixel were aggregated.

Statistical metrics used for evaluation include the correlation coefficient (r-value), mean bias, and ubRMSE. The ubRMSE, a modified version of traditional RMSE, is particularly effective in scenarios with systematic biases, such as those arising from spatial discrepancies between satellite and ground-based platforms [61]. This metric provides a clearer assessment of true error in satellite estimates by minimizing the influence of biases.

III. Results

A. Performance of Input Satellite SSM Products

To accurately evaluate the performance of the proposed module for enhancing the temporal resolution (or "temporal downscaling") of daily scale SSM, it is essential to first assess the accuracy of the input data being downscaled. In this context, the AMSR2 SSM at the 36-km scale and the spatially downscaled SSM (aggregated to 6-km grids), as shown in Fig. 2, were both validated using ground-based measurements over the corresponding spatial grids within the study area. The validation results are presented in Fig. 6(a) and (b). The original AMSR2 SSM data exhibited a ubRMSE of 0.047 cm^3/cm^3 and a bias of -0.043 cm^3/cm^3. In comparison, the spatially downscaled SSM at the 1-km scale demonstrated higher accuracy, with a ubRMSE of 0.029 cm^3/cm^3 and a bias of -0.018 cm^3/cm^3 over the course of 2019. However, the improved performance of the spatially downscaled SSM compared with the original AMSR2 SSM does not necessarily imply that the downscaling framework [32] increases the accuracy of the original microwave-based SSM estimates. This discrepancy may be attributed to the difference in the spatial distribution of ground soil moisture stations. As highlighted in Fig. 1, the ground stations within the entire 36-km grid are sparser than those within the selected 6-km subgrid. This disparity likely reduces the spatial correlation between satellite and ground-based data at the 36-km grid, which negatively impacts the apparent evaluation accuracy of the AMSR2 SSM data.

In addition, Fig. 6(c) shows that SMAP SSM, as the counterpart to AMSR2, achieves a lower ubRMSE of 0.037 cm^3/cm^3 and a higher correlation coefficient (r-value = 0.52) than AMSR2, albeit with a larger negative bias of approximately -0.053 cm^3/cm^3. This suggests that SMAP SSM slightly outperforms AMSR2 in certain aspects. However, the performance gap is relatively minor and does not indicate a significant inferiority of the selected AMSR2 SSM, especially when considering the smaller sample size of SMAP and the sparser distribution of ground validation sites. Furthermore, when evaluated alongside the spatially downscaled SSM—which demonstrates notably improved accuracy—a conservative yet robust conclusion can be drawn: the downscaled estimates derived from AMSR2 SSM achieve accuracy levels comparable to those of the SMAP benchmark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

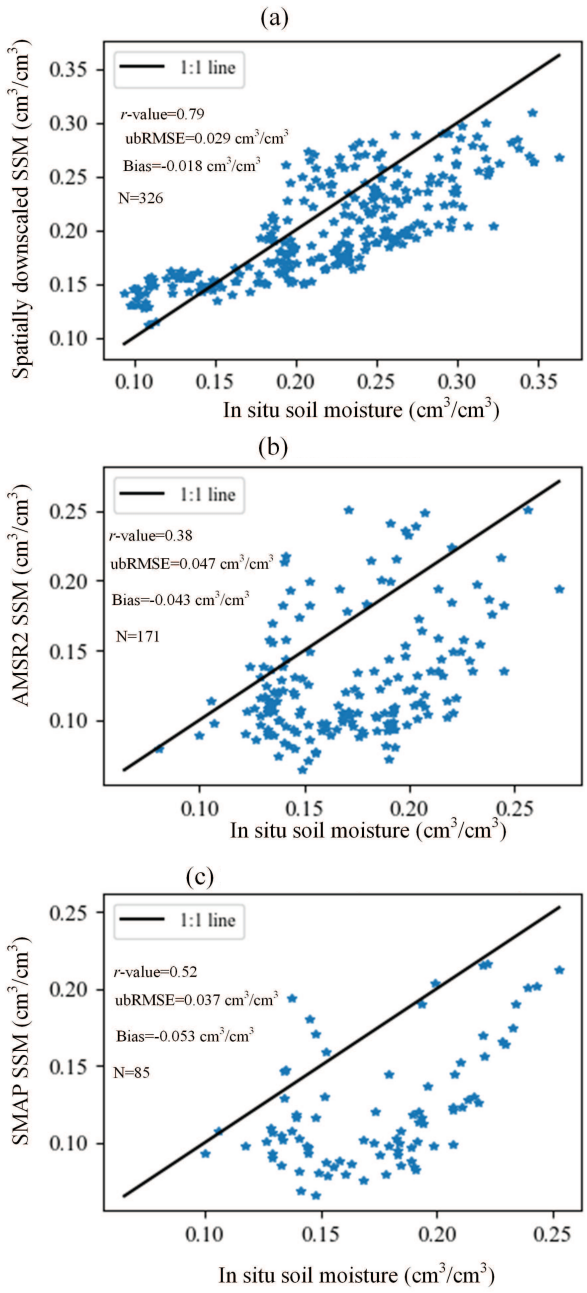


Fig. 6. Validation results for (a) 1-km spatially downscaled SSM (resampled at the 6-km scale), (b) 36-km AMSR2 SSM, and (c) 36-km SMAP SSM. The validations are made within the highlighted grids at 6-/36-km scales in our study area, as shown in Fig. 1, throughout the year of 2019.

Overall, the evaluation supports the conclusion that both the original AMSR2 SSM and the spatially downscaled SSM meet the basic accuracy standard of the advanced SMAP product (~0.04 cm^3/cm^3). This confirms that the input data are suitable for the temporal downscaling process.

B. Performance of Correlation Between SSMA and TIA

This section aims to assess the feasibility of (6), as proposed in Step (ii) of Section II-D. To achieve this, the potential linear correlations between FY-4A-based TIA and in situ SSMA were analyzed for each month in 2019. The results, displayed in Fig. 7, indicate that there is a statistically significant relationship (at $p < 0.05$) between these two variables across all months. Stronger correlations were observed during February to November, with the coefficient of determination (R^2) for the fitting of (6) ranging between 0.51 and 0.87. In contrast, the relatively lower R^2 values observed in the months of January and December suggest that the model may perform less effectively under colder temperature conditions.

Following this, the model was fit for each month using SSMA inputs derived from satellite observations at 1:30 A.M. on a daily scale (as outlined in Step (iii) of Section II-D). Scatter plots in Fig. 8 demonstrate the relationships between satellite-based SSMA and TIA. It is noteworthy that no results were provided for January, February, and December due to a lack of valid satellite SSM and FY-4A LST data pairs between 1:00 and 2:00 A.M. during these months. Since all model fits in Fig. 8 passed the significance test at $p < 0.05$ (with $R^2 > 0.40$), it is generally feasible to apply satellite daily SSM observations as inputs to (6), which was originally established at the site scale.

However, the R^2 values in Fig. 8 are generally lower than those in Fig. 7. To address this concern, we additionally calculated a complementary metric, referred to as the residual RMSE of SSMA, for each subplot. The residual RMSE is defined as the root mean square error between the observed SSMA values and the corresponding values estimated from the linear fit, i.e., $k_{\mathrm{SSMA-TIA}} \times \mathrm{TIA} + b_{\mathrm{SSMA-TIA}}$, where $k_{\mathrm{SSMA-TIA}}$ and $b_{\mathrm{SSMA-TIA}}$, respectively, denote the linear slope and intercept of the linear regression of TIA against SSMA in each subplot. As shown in Fig. 8, the residual RMSE remains below 0.02 cm^3/cm^3 across all months. Given that SSMA has the same unit as the actual SSM estimates, this error level, equivalent to half the standard SMAP product accuracy threshold (~0.04 cm^3/cm^3), provides strong support for the reliability of the established relationship in (6). The apparent inconsistency between the residual RMSE and the R^2 values in Fig. 8 is primarily due to the limited dynamic range of SSMA at the intraday scale. Because the analysis is restricted to the midnight time window, the x-axis range (SSMA) in each subplot of Fig. 8 is significantly narrow, with their maximum range below 0.1 cm^3/cm^3. This compressed range increases the sensitivity of the R^2 metric to even minor deviations in the y-axis variable (TIA), thereby diminishing the perceived strength of the linear correlation.

To further validate the robustness of (6), hourly scale remote sensing SSM outputs from this model were subsequently generated (following Steps (iii) and (iv) in Section II-D) and evaluated in Section III-C.

C. Performance of Estimated Hourly Scale SSM

All hourly scale SSM downscaled to the 6-km pixel resolution from March to November in 2019 are compared with in situ measurements, as shown in Fig. 9. The results demonstrate a ubRMSE of 0.036 cm^3/cm^3 and a mean bias of −0.019 cm^3/cm^3 for the spatiotemporally downscaled SSM, indicating that the downscaled data generally maintains the basic accuracy standard of the original 36-km SMAP data (~0.04 cm^3/cm^3).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

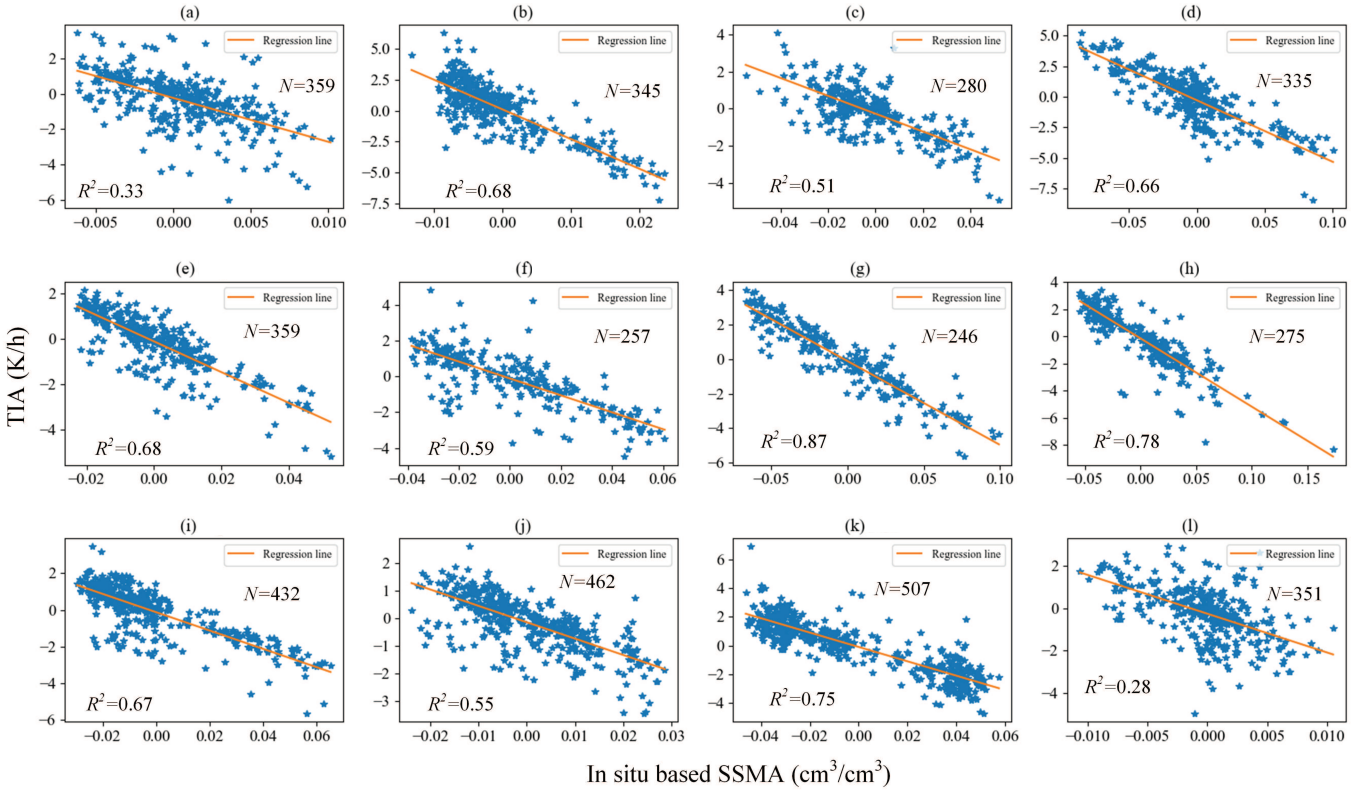


Fig. 7. Scatter plots for demonstrating linear correlation (6) between FY-4A-based LST and in situ-based SSMA (observed at all hourly scale moments) for each month in 2019. The correlations in all months have passed the significance test at $p < 0.05$. (a) January. (b) February. (c) March. (d) April. (e) May. (f) June. (g) July. (h) August. (i) September. (j) October. (k) November. (l) December.

In Fig. 10(a), we further evaluate the temporal consistency between the satellite-derived and ground-based soil moisture data. The time series over the study period reveals that the major seasonal cycles of the ground soil moisture data are well captured and reflected by the hourly scale satellite SSM estimates. However, significant divergence is observed at shorter temporal scales. To investigate this, an inset figure in Fig. 10(a) focuses on a typical time window between days of year (DOYs) 132 and 136 in 2019. The inset figure clearly shows that during these four days, the satellite SSM estimates exhibit much more pronounced fluctuations compared with the ground soil moisture measurements. Despite this, the bias values for data within each day remain within a relatively small range (< 0.03 cm^3/cm^3), suggesting that the apparent divergence at the intradaily scale is exaggerated.

To confirm this conclusion, we calculated the cumulative histograms for ubRMSE and bias values for each day during the study period, as shown in Fig. 10(b) and (c), respectively. The ubRMSE values are predominantly below 0.02 cm^3/cm^3, and the majority of the bias values fall within the range of $[-0.04, 0.04]$ cm^3/cm^3, consistent with the results from the inset figure in Fig. 10(a). Given that these values remain comparable to the basic accuracy standard of the SMAP SSM product, the estimation errors of the SSM data at the intradaily scale can be considered acceptable.

The time-series pattern of ground measurements reveals that soil moisture variability at the intradaily scale is inherently lower than at the seasonal scale. This suggests that our evaluation process is more sensitive to errors in satellite estimates at shorter temporal scales. This sensitivity also explains the exaggerated divergence between the time series of different data sources, as shown in the inset figure of Fig. 10(a). A potential solution to mitigate this divergence could involve applying a time-series filter to the hourly scale satellite SSM estimates within each diurnal cycle. Examples include the Wiener filter [62] or Whittaker filter [63], [64]. This approach would be particularly useful if the developed methodology is to be utilized for generating hourly SSM products in the future. However, it was not implemented in the current study, as our primary research objective was to evaluate the feasibility of the TIA-based theory for enhancing the frequency of satellite SSM independently, without the influence of additional procedures.

In this context, a more appropriate method for evaluating the intradaily time-series pattern of satellite SSM is to focus exclusively on dates when ground soil moisture exhibits significant variability. Following this rationale, we conducted the intradaily assessment only on specific dates characterized by notable fluctuations in ground soil moisture data, defined by intradaily standard deviations exceeding a threshold of 0.02 cm^3/cm^3. Based on this criterion, time-series data from ground measurements and satellite observations were selected for six representative dates, as depicted in Fig. 11. To further highlight the advantages of the proposed hourly SSM product in capturing subdaily soil moisture variations, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

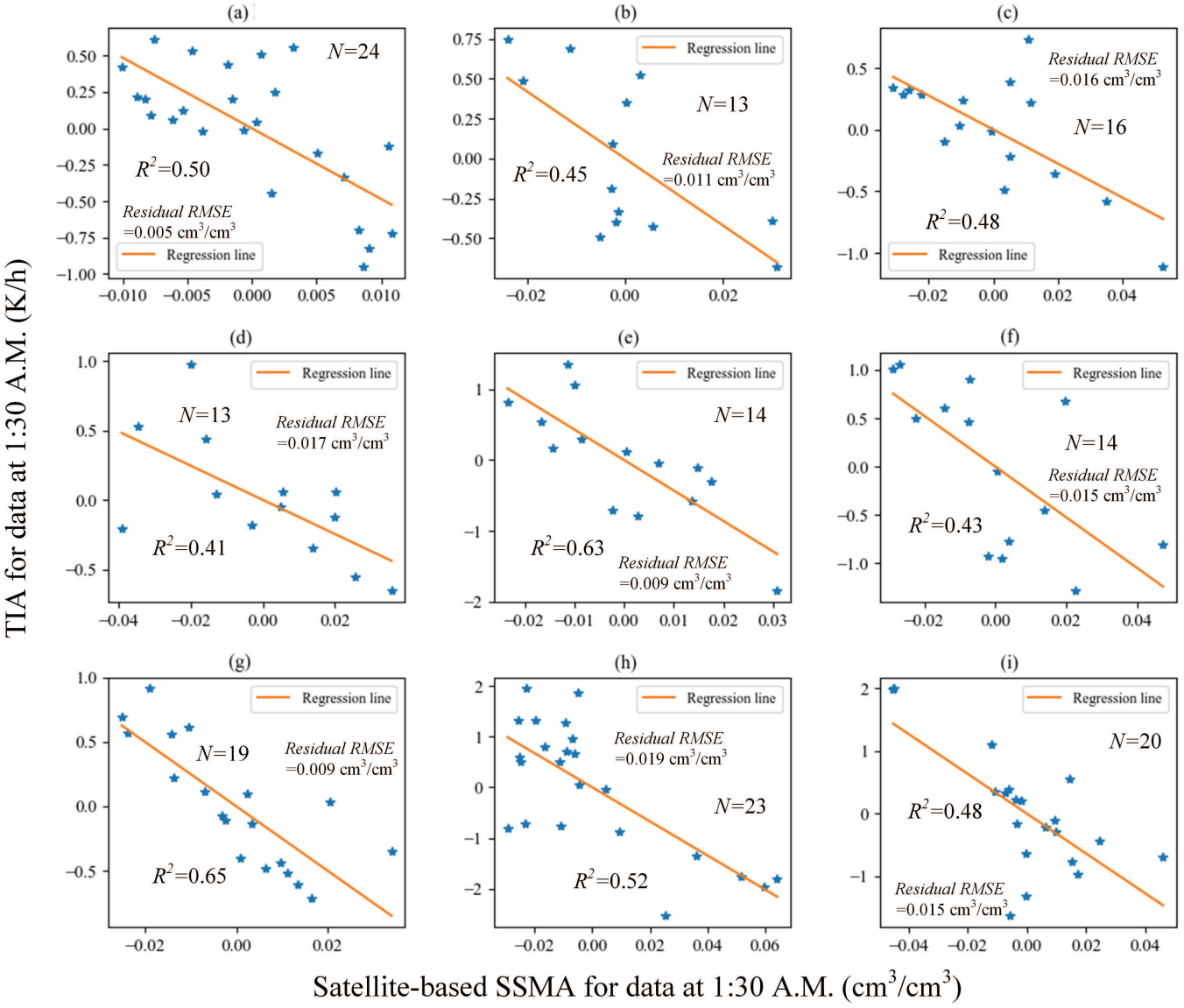


Fig. 8. Scatter plots for demonstrating linear correlation (6) between FY-4A-based LST and satellite-based SSMA (observed only at 1:30 A.M. of local time) from March to November in 2019. The correlations in all displayed months have passed the significance test at $p < 0.05$. (a) March. (b) April. (c) May. (d) June. (e) July. (f) August. (g) September. (h) October. (i) November.

also conducted a cross comparison with an existing subdaily product. As remotely sensed subdaily SSM products are scarce, we referred instead to the SMAP Level-4 product [65], which provides seamless 3-hourly estimates by assimilating SMAP Level-3 daily retrievals into NASA's Catchment Land Surface Model [66]. For consistency, the 9-km SMAP Level-4 product (DOI:https://dx.doi.org/10.5067/T5RUATAQREF8) was resampled to the target 6-km grid cell in our study area using bilinear interpolation. Each estimate represents the average soil moisture over a 3-h window, and the resulting time series was evaluated alongside the hourly retrievals from our algorithm in Fig. 11.

The analysis shows strong agreement between our satellite-based hourly SSM estimates and ground observations across all selected days. Correlation coefficients consistently exceeded 0.60, and ubRMSE values remained below 0.02 cm^3/cm^3. Although a negative mean bias was observed on April 26, the hourly retrievals still successfully reproduced the gradual increase in soil moisture captured by in situ data, particularly around 20:00 local time. This result demonstrates the capacity of the developed framework to track subdaily soil moisture dynamics despite occasional bias. By contrast, the SMAP Level-4 product exhibited limited variability throughout the diurnal cycle, failing to reproduce the pronounced subdaily fluctuations evident in the ground measurements. Despite its nominal 3-h resolution, the assimilation-based product appears to lack sensitivity to rapid changes in soil moisture. These findings underscore the added value of our framework, which complements existing products such as SMAP Level-4 by providing enhanced temporal resolution and improved sensitivity to intradaily dynamics.

In our final analysis, we utilized all data in the year to evaluate the validation statistics of hourly scale satellite SSM across different local times within a diurnal cycle,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

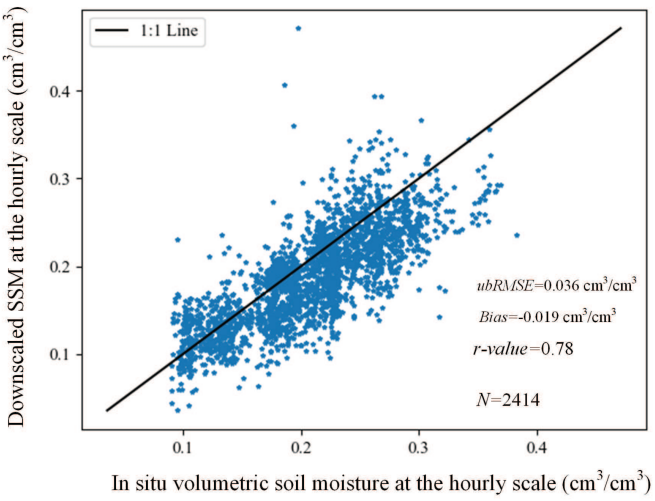


Fig. 9. Scatter plots for comparison between hourly scale SSM at 6 km and the spatiotemporally corresponsive ground measurements for all dates in 2019.

TABLE I

GROUND SOIL MOISTURE NETWORKS WIDELY USED IN PREVIOUS EVALUATIONS OF SATELLITE SSM PRODUCTS ACROSS KEY FY-4A OBSERVATIONAL FOOTPRINTS IN THE ASIA–PACIFIC REGION, ALONG WITH THE PRIMARY REASONS WHY NONE OF THEM WERE EMPLOYED IN THIS STUDY

Network name	Active observational period	Primary reason for not being employed by the current study	Reference
Naqu	2010.1-2019.12	No dates simultaneously satisfy both conditions: intraday standard deviation > 0.02 cm³/cm³ and ≥10 clear-sky hourly LST observations	Su et al. [67]
Maqu	2010.1-2019.12	(Same to Naqu)	
Ngari	2010.1-2019.12	Site density does not meet the requirement of ≥4 stations within a single 6-km pixel	Su et al. [68]
Wudaoliang	2020.1-2022.8	(Same to Ngari)	Zheng et al. [69]
Yanco	2021.1-2022.12	No dates satisfy the condition of intraday standard deviation > 0.02 cm³/cm³	Panciera et al. [70]

as shown in Fig. 12. The results indicate that relatively poorer performance was observed during the local times of 2:00–4:00, 12:00–14:00, and 18:00–20:00, where ubRMSE values exceeded 0.04 cm^3/cm^3 and r-values fell below 0.8. This discrepancy is likely attributable to the relatively flat slope of the DTC function during these periods (referring to Fig. 3). According to (3), the slope of the DTC is critical for estimating TIA, which is essential for retrieving hourly scale satellite SSM. A flat slope during certain times suggests a lower sensitivity of the thermal inertia-based TIA in capturing variations in SSM, thereby diminishing the effectiveness of (6) during these intervals. Notably, while other times, such as 20:00–22:00 and 0:00–2:00, also exhibit flat slopes in the DTC, the proximity of these moments to the spatially downscaled SSM inputs (at 1:30 local time) means that the SSMA(t_0) component in (5) tends to be quite similar in most cases. This similarity helps mitigate the uncertainty arising from the flat slope of the DTC, allowing for more reliable estimates of SSM during those hours.

IV. DISCUSSION

A. Potential Universality and Limitations of the Proposed Methodological Framework on Regional Scales

Using solely satellite-based inputs, our study has successfully generated hourly scale satellite-based SSM data with overall acceptable accuracy (ubRMSE ≤ 0.04 cm^3/cm^3) within a pixel-scale region characterized by a dense network of ground soil moisture stations. The relatively homogeneous grassland land cover in this region suggests that the proposed framework holds potential for broader applicability across similarly homogeneous landscapes with moderate to low vegetation cover. This also aligns with the baseline validation standards adopted for SMAP SSM products. However, further efforts are necessary to assess the universality of the proposed methodology across wider spatial domains. The ISMN [47] provides access to a wide range of in situ soil moisture measurements, which support the validation of satellite-based SSM products at typical microwave pixel scales (≥25 km). In addition to the SMN-SDR sites employed in this study, several other ground networks have been widely used in previous evaluations of satellite SSM products across key FY-4A observational footprints in the Asia–Pacific region. These include the sites located in Naqu, Maqu, and Ngari [67], [68], the Wudaoliang sites [69], as well as the Yanco sites [70]. Due to limitations such as insufficient diurnal variation in soil moisture or inadequate spatial sampling density, these networks were not employed in the present analysis, with the detailed reasons summarized in Table I. This highlights the need for future studies to re-evaluate the feasibility and generalizability of the proposed framework using additional high-density, ground observation sites characterized by high-frequency soil moisture dynamics.

Another significant limitation of the current study is that the geostationary satellite-observed TIR LST input data are affected by cloud cover. In Fig. 13, we illustrate the number percentage of clear-sky LST observations available from the FY-4A AGRI platform for each date during the study period. It is evident that only a few dates meet the ideal condition of 100% clear-sky observations. The missing rate is particularly high during June and July (DOYs 160–200). This interference severely disrupts the continuity of the derived hourly scale SSM outputs. In addition, discontinuities in LST inputs can compromise the reliability of the DTC modeling process, leading to decreased accuracy in TIA. As our current study has not produced hourly scale soil moisture data during cloudy or rainy conditions, the applicability of this dataset is limited in monitoring changes in the underlying surface during extreme weather events, such as rapid storms or heavy rainfall.

Moreover, as illustrated in Fig. 12, hourly scale SSM estimates around midday or sunset are generally less accurate compared with those obtained at other times. Although this negative impact appears less significant in the current

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

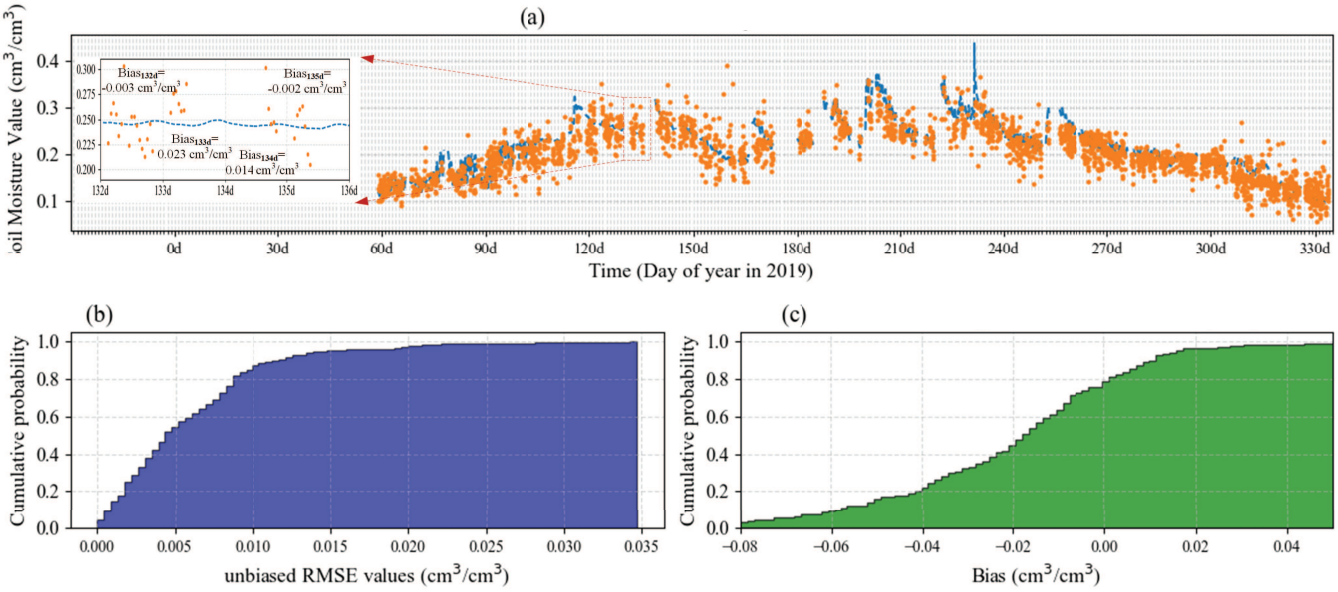


Fig. 10. (a) Time series of hourly soil moisture from ground measurements and satellite estimates in 2019. January, February, and December are excluded due to a lack of valid satellite SSM and FY-4A LST data pairs between 1:00 and 2:00 A.M. Inset: magnified view of the time series from days 132 to 136 of the year. (b) Cumulative histogram of ubRMSE for satellite estimates (compared with ground measurements) for each day of the year [corresponding to Fig. 10(a)]. (c) Cumulative histogram of bias for satellite estimates (compared with ground measurements) for each day of the year [corresponding to Fig. 10(a)].

experimental site, the inconsistency in accuracy among different time points may distort the intradaily time-series pattern when applied at regional scales. Addressing these limitations will be crucial for enhancing the robustness and applicability of the methodology across diverse environmental conditions.

Finally, beyond the uncertainties associated with the temporal downscaling submodule, the derived hourly SSM may also inherit errors propagated from the spatially downscaled 1-km daily SSM product generated in the first stage of the framework. Although our evaluation (see Fig. 6) indicates that such propagated uncertainties are minimal within the present study area, they warrant closer examination in broader applications. As shown in our previous work on 1-km daily SSM generation [32], spatial downscaling tends to become less reliable under cloudy or rainy conditions, in regions with complex topography, or in areas with dense vegetation cover. Consequently, further improvements to the first-stage spatial downscaling submodule will be essential to mitigate uncertainty transmission and, ultimately, to enhance the robustness of the final hourly SSM estimates.

B. Implications and Future Improvement of the Proposed Algorithm

The proposed methodological framework pioneers the temporal downscaling of polar-orbiting satellite-based SSM products from daily to hourly scales by integrating TIR observations from geostationary platforms. In this framework, the spatial resolution of the final hourly scale SSM dataset is constrained by that of the geostationary observations—for example, the 6-km resolution used in this study, aligned with FY-4A data. While such spatial resolution may be considered insufficient for certain high-precision hydrological or agricultural applications, the true value of the framework lies in its demonstration of how geostationary TIR observations can effectively capture subdaily SSM dynamics. With the anticipated advancement of geostationary satellite observation systems in the future, this framework is expected to play an increasingly important role in producing hourly SSM products with enhanced spatial resolution.

In addition, we propose several academic recommendations to further improve the framework for large-scale applications.

1) Improvement on Continuity of Intradaily Time Series: To ensure high continuity of hourly scale SSM outcomes, it is essential to utilize high-quality, all-weather hourly LST products. Recent advancements have led to the development of such products based on geostationary satellite platforms [71], [72], [73]. Future experiments should assess the feasibility of these existing products as inputs to our models. It's also worth noting that realistic LST signals under cloudy and clear conditions exhibit different response patterns on SSM [32], [74]. Therefore, additional investigation is warranted to examine whether clear-sky-like LST reconstructions or actual cloudy-sky LST signals provide a more suitable basis for temporally downscaling SSM during rapidly changing weather within a diurnal cycle.

In addition, as demonstrated in Fig. 11, the SMAP Level-4 product, although provided at a nominal 3-h resolution, shows little sensitivity to subdaily soil moisture fluctuations, in contrast to our framework that better captures diurnal variability. This highlights the potential of our satellite-based method to complement model-driven products. While our algorithm currently relies on clear-sky LST inputs and therefore cannot provide continuous estimates on its own, assimilating its hourly SSM outputs into land surface models (e.g., Catchment) could substantially enhance their ability to reproduce true subdaily hydrological dynamics. Such integration would not

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

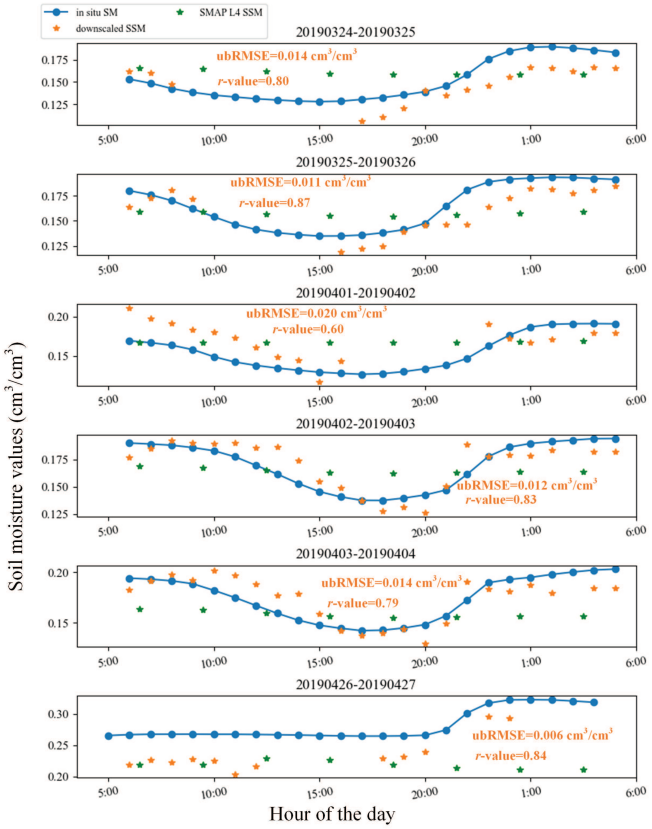


Fig. 11. Intradaily time series of soil moisture at the hourly scale for six representative dates, showing downscaled satellite SSM (6 km), in situ ground measurements, and the SMAP Level-4 SSM product (resampled from 9 to 6 km). The selected dates were chosen based on significant intradaily variability in ground soil moisture, defined by a standard deviation exceeding 0.02 cm^3/cm^3. Statistical metrics (ubRMSE and *r*-values) are reported only for the comparison between our developed hourly satellite SSM and ground observations in each subfigure. For the SMAP Level-4 SSM, no such metrics are provided because its time series exhibits minimal variability, leading to an apparent lack of correlation with ground measurements.

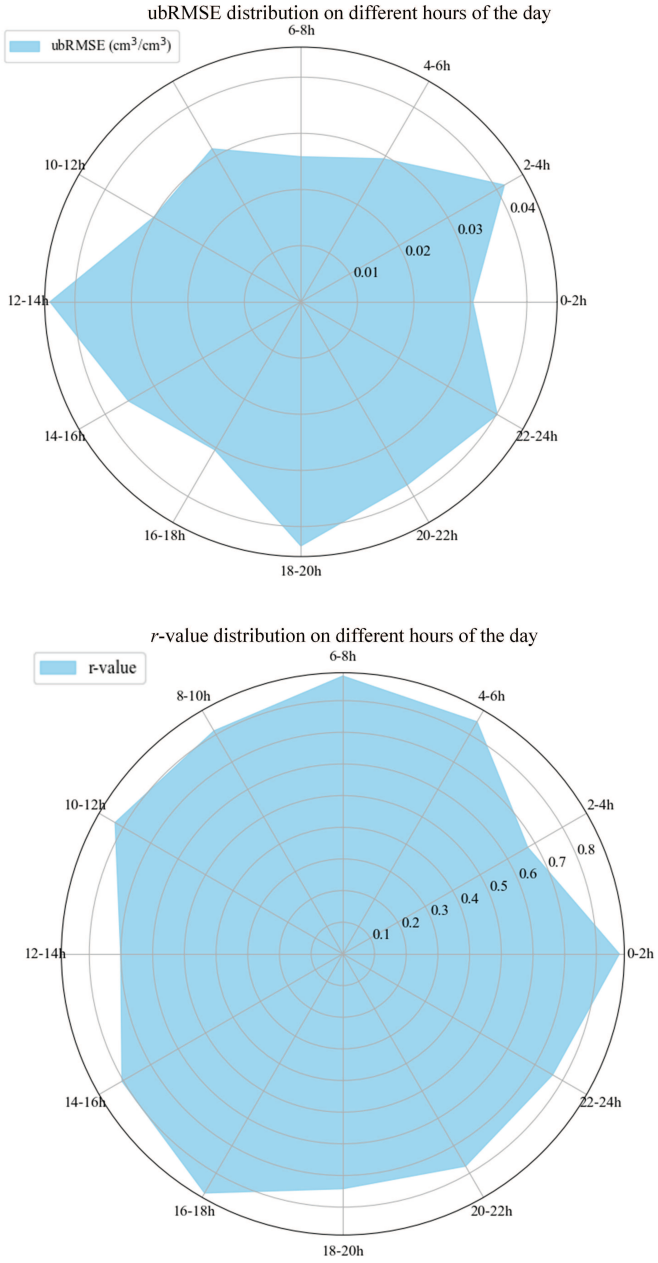


Fig. 12. Distribution of ubRMSE and *r*-values with respect to different hours of local time within a diurnal cycle, by validating the hourly scale 6-km satellite SSM against ground soil moisture measurements throughout the year of 2019.

only improve the temporal continuity of our retrievals but also strengthen model-based estimates of soil moisture and related environmental variables under rapidly changing surface conditions.

2) Improvement in Accuracy and Consistency at Different Local Times: As shown in Fig. 12, discrepancies in DTC slopes within a diurnal cycle have led to inconsistent SSM accuracy at different times, particularly around mid-noon and sunset. However, the negative impact of flat DTC slopes can be mitigated if the observational times of the hourly scale SSM output and daily scale input are closely aligned. To address this, we recommend implementing the "temporal downscaling" model using spatially downscaled SSM data with a higher frequency than a single observation per day (currently at 1:30 A.M.). Global microwave SSM products are available at various local times, such as dawn–dusk (~6:00 A.M./P.M., e.g., SMAP), early afternoon (~1:30 A.M./P.M., e.g., AMSR2), and middle morning (~10:00 A.M./P.M., e.g., ASCAT). By applying "Submodule 1" (see Fig. 2) on a fused dataset of these microwave SSM products, we can potentially produce SSM datasets with up to six observations in a diurnal cycle, which would enhance the accuracy of the hourly output from "submodule 2" around mid-noon and sunset.

3) Improvement on the Evaluation of the Algorithm Over Different Regions of the World: The effectiveness of satellite remote sensing models is inherently tied to the quality of the benchmark datasets used for validation. As discussed above, our current evaluation, confined to a single 6-km pixel, is limited in scope, while comparable validation networks in other regions fall short of the necessary criteria for this study. Future work would greatly benefit from expanded ground-based monitoring networks with higher station density, particularly in regions exhibiting frequent and dynamic soil moisture

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

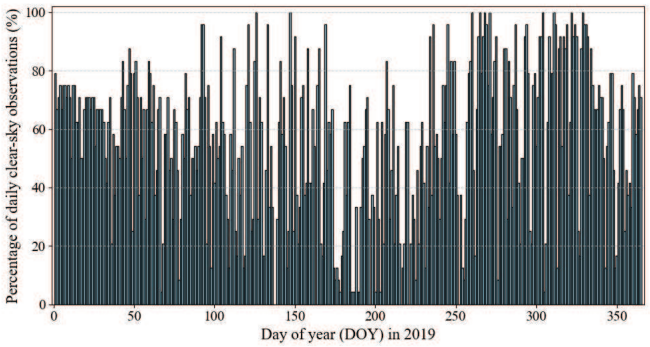


Fig. 13. Number percentage (%) of clear-sky LST observations from FY-4A AGRI for each date in 2019.

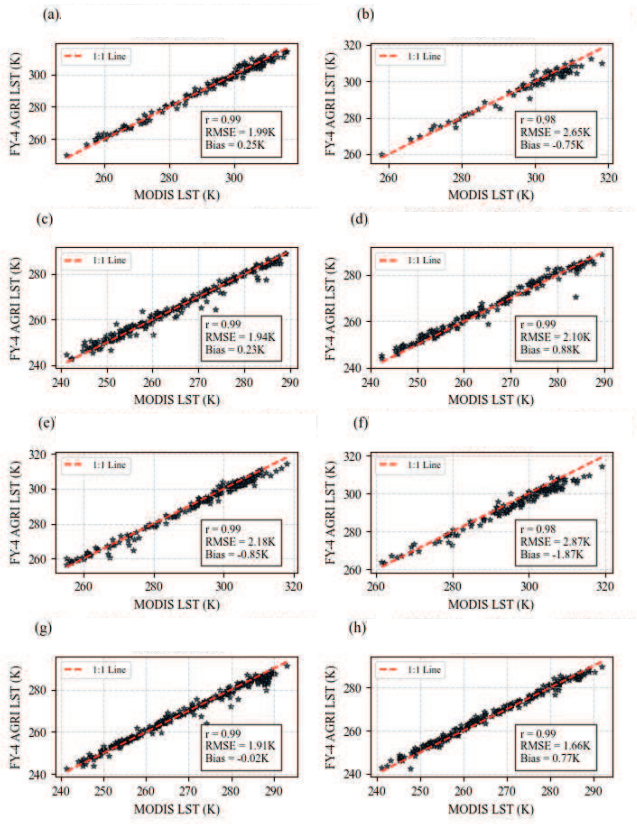


Fig. 14. Validation of FY-4A AGRI LST using MODIS LST as the benchmark. Results are shown for four local moments (10:30 A.M./P.M. and 1:30 A.M./P.M.) at 36- and 6-km grids in the study area, demonstrating strong agreement between the datasets. (a) Aqua MODIS-based daytime (1:30 P.M.), mode at 36 km. (b) Aqua MODIS-based daytime (1:30 P.M.), mode at 6 km. (c) Aqua MODIS-based nighttime (1:30 A.M.), mode at 36 km. (d) Aqua MODIS-based nighttime (1:30 A.M.), mode at 6 km. (e) Terra MODIS-based daytime (10:30 A.M.), mode at 36 km. (f) Terra MODIS-based daytime (10:30 A.M.), mode at 6 km. (g) Terra MODIS-based nighttime (10:30 P.M.), mode at 36 km. (h) Terra MODIS-based nighttime (10:30 P.M.), mode at 6 km.

fluctuations. Future work would greatly benefit from expanded ground-based monitoring networks with higher station density, particularly in regions exhibiting frequent and dynamic soil moisture fluctuations. Such environments would enable a more robust assessment of the algorithm's capacity to resolve short-term variations in SSM. Furthermore, incorporating LST data from other geostationary satellite platforms—such as the Meteosat Second Generation (MSG) over Europe and Africa or the Geostationary Operational Environmental Satellite (GOES) over the Americas [75], would facilitate broader testing of the algorithm across diverse geographic and climatic settings, ultimately advancing its global applicability.

V. Conclusion

In this study, we proposed an innovative methodological framework designed to produce high-resolution SSM data with an hourly revisit cycle, a feature that has received limited attention in prior research. Central to this framework is a novel model that leverages the high-frequency capabilities of geostationary satellite observations to enhance the temporal resolution of existing polar-orbit satellite-based SSM products. Implemented within a well-established ground network for soil moisture validation in North China, our approach yielded hourly SSM time-series data at a 6-km resolution. Validation results indicate significant improvements in both spatial and temporal resolutions compared with conventional 36-km and daily scale SMAP (or SMAP-like) microwave SSM products, while achieving a comparable accuracy (ubRMSE ≤ 0.04 cm^3/cm^3). Nonetheless, further research is essential to enhance the "all-weather continuity" of the time-series outputs, as the current framework is limited to clear weather conditions. In addition, extensive evaluations are required to assess the framework's applicability across diverse geographical regions, which is crucial for scaling the methodology to generate regional-scale hourly SSM data. This advancement has the potential to significantly benefit sectors such as meteorology, hydrology, and agriculture by providing timely and accurate soil moisture information.

Appendix

To further corroborate the reliability of the FY-4A AGRI LST dataset utilized in this study, a targeted validation was conducted for the two study pixels with spatial resolutions of 36 and 6 km, respectively. The validation employed the 1-km daily MODIS LST products (MOD11A1 and MYD11A1, Version 6.1) as benchmark data for the year 2019. The 1-km MODIS LST retrievals flagged as "good quality" were aggregated to match the spatial resolutions of the 36- and 6-km grids. Temporally, the FY-4A AGRI LST observations were averaged to align with the MODIS local overpass moments: for MOD11A1 (Terra MODIS, ~10:30 A.M./P.M.), FY-4A observations at 10:15, 10:30, and 10:45 A.M./P.M. were used; for MYD11A1 (Aqua MODIS, ~1:30 A.M./P.M.), observations at 1:15, 1:30, and 1:45 A.M./P.M. were used. This ensured a direct comparison between the datasets at both spatial and temporal scales.

The validation results, as illustrated in Fig. 14, demonstrate a high level of consistency between the FY-4A AGRI LST and MODIS LST datasets. All scenarios achieved exceptionally high correlation coefficients, reaching up to 0.98 and indicating strong agreement. The RMSE values for the 6-km pixel were consistently below 3 K, while those for the 36-km pixel were even lower, averaging around 2 K. In most cases, the bias

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

values fell within the range of [−1, 1] K, with the only exception being the 10:30 A.M. validation for the 6-km pixel, where the absolute bias value slightly increased to 1.87 K.

Overall, these results highlight the robustness of the FY-4A AGRI LST product for LST monitoring and DTC modeling. The strong agreement with the widely recognized MODIS LST datasets underscores the suitability of the FY-4A AGRI LST product for accurate and consistent LST analysis at both the diurnal and the seasonal cycles. This supports the validity of using the dataset for hourly SSM estimation in the current study.

ACKNOWLEDGMENT

The authors thank editors and anonymous reviewers for their critical but constructive suggestions.

REFERENCES

[1] L. Brocca et al., "Improving runoff prediction through the assimilation of the ASCAT soil moisture product," *Hydrol. Earth Syst. Sci.*, vol. 14, no. 10, pp. 1881–1893, Oct. 2010.

[2] R. D. Koster et al., "Regions of strong coupling between soil moisture and precipitation," *Science*, vol. 305, no. 5687, pp. 1138–1140, Aug. 2004.

[3] D. Yin et al., "Understanding the role of initial soil moisture and precipitation magnitude in flood forecast using a hydrometeorological modelling system," *Hydrological Processes*, vol. 36, no. 10, p. 14710, Oct. 2022.

[4] N. K. Singh, R. E. Emanuel, B. L. McGlynn, and C. F. Miniat, "Soil moisture responses to rainfall: Implications for runoff generation," *Water Resour. Res.*, vol. 57, no. 9, p. 2020, Sep. 2021.

[5] A. K. Nayak, B. Biswal, and K. P. Sudheer, "Role of hydrological model structure in the assimilation of soil moisture for streamflow prediction," *J. Hydrol.*, vol. 598, Jul. 2021, Art. no. 126465.

[6] J. Muñoz-Sabater et al., "Assimilation of SMOS brightness temperatures in the ECMWF integrated forecasting system," *Quart. J. Roy. Meteorological Soc.*, vol. 145, no. 723, pp. 2524–2548, Jul. 2019.

[7] N. Rodríguez-Fernández et al., "SMOS neural network soil moisture data assimilation in a land surface model and atmospheric impact," *Remote Sens.*, vol. 11, no. 11, p. 1334, Jun. 2019.

[8] A. K. Singh, "Precision farming," Water Technology Centre, New Delhi, India, Tech. Rep., 2010, pp. 165–174. [Online]. Available: https://www.academia.edu/download/60139936/14-Precision_Farming_Lecture20190728-38693-ymmq9k.pdf

[9] J. D. Bolten, W. T. Crow, X. Zhan, T. J. Jackson, and C. A. Reynolds, "Evaluating the utility of remotely sensed soil moisture retrievals for operational agricultural drought monitoring," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 3, no. 1, pp. 57–66, Mar. 2010.

[10] Y. Inoue et al., "Season-long daily measurements of multifrequency (KA, Ku, X, C, and L) and full-polarization backscatter signatures over paddy Rice field and their relationship with biological variables," *Remote Sens. Environ.*, vol. 81, nos. 2–3, pp. 194–204, Aug. 2002.

[11] J. W. Godt, R. L. Baum, and A. F. Chleborad, "Rainfall characteristics for shallow landsliding in Seattle, Washington, USA," *Earth Surf. Processes Landforms*, vol. 31, no. 1, pp. 97–110, Jan. 2006.

[12] L. Brocca, F. Ponziani, T. Moramarco, F. Melone, N. Berni, and W. Wagner, "Improving landslide forecasting using ASCAT-derived soil moisture data: A case study of the Torgiovannetto landslide in central Italy," *Remote Sens.*, vol. 4, no. 5, pp. 1232–1244, May 2012.

[13] E. G. Njoku and D. Entekhabi, "Passive microwave remote sensing of soil moisture," *J. Hydrol.*, vol. 184, nos. 1–2, pp. 101–129, Oct. 1996, doi: 10.1016/0022-1694(95)02970-2.

[14] E. G. Njoku, "Surface temperature estimation over land using satellite microwave radiometry," in *Passive Microwave Remote Sensing of Land–Atmosphere Interactions*. Boca Raton, FL, USA: CRC Press, 1995, pp. 509–530.

[15] A. G. Konings et al., "Detecting forest response to droughts with global observations of vegetation water content," *Global Change Biol.*, vol. 27, no. 23, pp. 6005–6024, Dec. 2021.

[16] R. M. Parinussa, T. R. H. Holmes, N. Wanders, W. A. Dorigo, and R. A. M. de Jeu, "A preliminary study toward consistent soil moisture from AMSR2," *J. Hydrometeorol.*, vol. 16, no. 2, pp. 932–947, Apr. 2015, doi: 10.1175/jhm-d-13-0200.1.

[17] D. Entekhabi et al., "The soil moisture active passive (SMAP) mission," *Proc. IEEE*, vol. 98, no. 5, pp. 704–716, May 2010, doi: 10.1109/JPROC.2010.2043918.

[18] Y. H. Kerr et al., "The SMOS soil moisture retrieval algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 5, pp. 1384–1403, May 2012.

[19] W. Wagner et al., "The ASCAT soil moisture product: A review of its specifications, validation results, and emerging applications," *Meteorologische Zeitschrift*, vol. 22, no. 1, pp. 5–33, Feb. 2013.

[20] P. Song et al., "Soil moisture estimation based on FY-3E backscattering data for enhanced daily coverage to SMAP observations in the dawn-dusk orbit," *Remote Sens. Environ.*, vol. 309, Aug. 2024, Art. no. 114209.

[21] L. Zhu, S. Yuan, Y. Liu, C. Chen, and J. P. Walker, "Time series soil moisture retrieval from SAR data: Multi-temporal constraints and a global validation," *Remote Sens. Environ.*, vol. 287, Mar. 2023, Art. no. 113466.

[22] M. Khazaei, S. Hamzeh, N. N. Samani, A. Muhuri, K. Goïta, and Q. Weng, "A Web-based system for satellite-based high-resolution global soil moisture maps," *Comput. Geosci.*, vol. 170, Jan. 2023, Art. no. 105250.

[23] L. Brocca et al., "How much water is used for irrigation? A new approach exploiting coarse resolution satellite soil moisture products," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 73, pp. 752–766, Dec. 2018.

[24] D. J. Mulla, "Twenty five years of remote sensing in precision agriculture: Key advances and remaining knowledge gaps," *Biosystems Eng.*, vol. 114, no. 4, pp. 358–371, Apr. 2013, doi: 10.1016/j.biosystemseng.2012.08.009.

[25] O. Merlin, J. P. Walker, A. Chehbouni, and Y. Kerr, "Towards deterministic downscaling of SMOS soil moisture using MODIS derived soil evaporative efficiency," *Remote Sens. Environ.*, vol. 112, no. 10, pp. 3935–3946, Oct. 2008, doi: 10.1016/j.rse.2008.06.012.

[26] J. Peng, A. Loew, S. Zhang, J. Wang, and J. Niesel, "Spatial downscaling of satellite soil moisture data using a vegetation temperature condition index," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 558–566, Jan. 2016, doi: 10.1109/TGRS.2015.2462074.

[27] M. Piles et al., "A downscaling approach for SMOS land observations: Evaluation of high-resolution soil moisture maps over the Iberian Peninsula," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 9, pp. 3845–3857, Sep. 2014, doi: 10.1109/JSTARS.2014.2325398.

[28] P. Song, Y. Zhang, and J. Tian, "Improving surface soil moisture estimates in humid regions by an enhanced remote sensing technique," *Geophys. Res. Lett.*, vol. 48, no. 5, p. 2020, Mar. 2021, doi: 10.1029/2020gl091459.

[29] B. Fang, V. Lakshmi, R. Bindlish, and T. Jackson, "AMSR2 soil moisture downscaling using temperature and vegetation data," *Remote Sens.*, vol. 10, no. 10, p. 1575, Oct. 2018.

[30] N. N. Das et al., "The SMAP mission combined active-passive soil moisture product at 9 km and 3 km spatial resolutions," *Remote Sens. Environ.*, vol. 211, pp. 204–217, Jun. 2018.

[31] B. Fang, V. Lakshmi, M. Cosh, P.-W. Liu, R. Bindlish, and T. J. Jackson, "A global 1-km downscaled SMAP soil moisture product based on thermal inertia theory," *Vadose Zone J.*, vol. 21, no. 2, p. 20182, Mar. 2022, doi: 10.1002/vzj2.20182.

[32] P. Song, Y. Zhang, J. Guo, J. Shi, T. Zhao, and B. Tong, "A 1 km daily surface soil moisture dataset of enhanced coverage under all-weather conditions over China in 2003–2019," *Earth Syst. Sci. Data*, vol. 14, no. 6, pp. 2613–2637, Jun. 2022.

[33] N. N. Das et al., "The SMAP and Copernicus sentinel 1A/B microwave active-passive high resolution surface soil moisture product," *Remote Sens. Environ.*, vol. 233, Nov. 2019, Art. no. 111380, doi: 10.1016/j.rse.2019.111380.

[34] J. Peng et al., "A roadmap for high-resolution satellite soil moisture applications–confronting product characteristics with user requirements," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112162, doi: 10.1016/j.rse.2020.112162.

[35] T. Yang, J. Wang, Z. Sun, and S. Li, "Daily soil moisture retrieval by fusing CYGNSS and multi-source auxiliary data using machine learning methods," *Sensors*, vol. 23, no. 22, p. 9066, Nov. 2023.

[36] M. P. Clarizia, N. Pierdicca, F. Costantini, and N. Floury, "Analysis of CYGNSS data for soil moisture retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2227–2235, Jul. 2019.

[37] S. Jin et al., "Remote sensing and its applications using GNSS reflected signals: Advances and prospects," *Satell. Navigat.*, vol. 5, no. 1, p. 19, Dec. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.

[38] H. H. Nguyen et al., "From theory to hydrological practice: Leveraging CYGNSS data over seven years for advanced soil moisture monitoring," *Remote Sens. Environ.*, vol. 316, Jan. 2025, Art. no. 114509.
[39] Q. Yan, W. Huang, S. Jin, and Y. Jia, "Pan-tropical soil moisture mapping based on a three-layer model from CYGNSS GNSS-R data," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111944.
[40] T. Yang, W. Wan, J. Wang, B. Liu, and Z. Sun, "A physics-based algorithm to couple CYGNSS surface reflectivity and SMAP brightness temperature estimates for accurate soil moisture retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409715.
[41] F. Lei et al., "Quasi-global machine learning-based soil moisture estimates at high spatio-temporal scales using CYGNSS and SMAP observations," *Remote Sens. Environ.*, vol. 276, Jul. 2022, Art. no. 113041.
[42] W. Yang, F. Guo, X. Zhang, Y. Zhu, Z. Li, and Z. Zhang, "First quasi-global soil moisture retrieval using Fengyun-3 GNSS-R constellation observations," *Remote Sens. Environ.*, vol. 321, May 2025, Art. no. 114653, doi: 10.1016/j.rse.2025.114653.
[43] S. E. Hobbs et al., "G-CLASS: Geosynchronous radar for water cycle science–orbit selection and system design," *J. Eng.*, vol. 2019, no. 21, pp. 7534–7537, Nov. 2019.
[44] L. Kilic et al., "Expected performances of the Copernicus imaging microwave radiometer (CIMR) for an all-weather and high spatial resolution estimation of ocean and sea ice parameters," *J. Geophys. Res., Oceans*, vol. 123, no. 10, pp. 7564–7580, Oct. 2018, doi: 10.1029/2018jc014408.
[45] J. Zheng et al., "Assessment of 24 soil moisture datasets using a new in situ network in the Shandian river basin of China," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112891, doi: 10.1016/j.rse.2022.112891.
[46] T. Zhao et al., "Soil moisture experiment in the Luan river supporting new satellite mission opportunities," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111680.
[47] W. Dorigo et al., "The international soil moisture network: Serving Earth system science for over a decade," *Hydrol. Earth Syst. Sci. Discuss.*, vol. 25, no. 11, pp. 5749–5804, 2021.
[48] P. Yao et al., "A long term global daily soil moisture dataset derived from AMSR-E and AMSR2 (2002–2019)," *Scientific Data*, vol. 8, no. 1, p. 143, May 2021, doi: 10.1038/s41597-021-00925-8.
[49] D. Xian, P. Zhang, L. Gao, R. Sun, H. Zhang, and X. Jia, "Fengyun meteorological satellite products for Earth system science applications," *Adv. Atmos. Sci.*, vol. 38, no. 8, pp. 1267–1284, Aug. 2021, doi: 10.1007/s00376-021-0425-3.
[50] L. Dong, S. Tang, F. Wang, M. Cosh, X. Li, and M. Min, "Inversion and validation of FY-4A official Land Surface Temperature product," *Remote Sens.*, vol. 15, no. 9, p. 2437, May 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/9/2437
[51] O. Merlin, A. Al Bitar, J. P. Walker, and Y. Kerr, "An improved algorithm for disaggregating microwave-derived soil moisture based on red, near-infrared and thermal-infrared data," *Remote Sens. Environ.*, vol. 114, no. 10, pp. 2305–2316, Oct. 2010, doi: 10.1016/j.rse.2010.05.007.
[52] B. Molero et al., "SMOS disaggregated soil moisture product at 1 km resolution: Processor overview and first validation results," *Remote Sens. Environ.*, vol. 180, pp. 361–376, Jul. 2016, doi: 10.1016/j.rse.2016.02.045.
[53] T. N. Carlson, R. R. Gillies, and E. M. Perry, "A method to make use of thermal infrared temperature and NDVI measurements to infer surface soil water content and fractional vegetation cover," *Remote Sens. Rev.*, vol. 9, nos. 1–2, pp. 161–173, Mar. 1994.
[54] B. Fang and V. Lakshmi, "AMSR-E soil moisture disaggregation using MODIS and NLDAS data," in *Remote Sensing of the Terrestrial Water Cycle*. Hoboken, NJ, USA: Wiley, 2014, pp. 277–304.
[55] R. M. Parinussa, T. R. H. Holmes, M. T. Yilmaz, and W. T. Crow, "The impact of Land Surface Temperature on soil moisture anomaly detection from passive microwave observations," *Hydrol. Earth Syst. Sci.*, vol. 15, no. 10, pp. 3135–3151, Oct. 2011, doi: 10.5194/hess-15-3135-2011.
[56] Y. Mao, Z. Wu, H. He, G. Lu, H. Xu, and Q. Lin, "Spatio-temporal analysis of drought in a typical plain region based on the soil moisture anomaly percentage index," *Sci. Total Environ.*, vol. 576, pp. 752–765, Jan. 2017, doi: 10.1016/j.scitotenv.2016.10.116.
[57] C. F. Ropelewski and M. S. Halpert, "Precipitation patterns associated with the high index phase of the southern oscillation," *J. Climate*, vol. 2, no. 3, pp. 268–284, Mar. 1989.
[58] T. R. H. Holmes, W. T. Crow, M. Tugrul Yilmaz, T. J. Jackson, and J. B. Basara, "Enhancing model-based Land Surface Temperature estimates using multiplatform microwave observations," *J. Geophys. Res., Atmos.*, vol. 118, no. 2, pp. 577–591, Jan. 2013, doi: 10.1002/jgrd.50113.
[59] T. R. H. Holmes, W. T. Crow, C. Hain, M. C. Anderson, and W. P. Kustas, "Diurnal temperature cycle as observed by thermal infrared and microwave radiometers," *Remote Sens. Environ.*, vol. 158, pp. 110–125, Mar. 2015.
[60] B. Fang, V. Lakshmi, R. Bindlish, T. J. Jackson, M. Cosh, and J. Basara, "Passive microwave soil moisture downscaling using vegetation index and skin surface temperature," *Vadose Zone J.*, vol. 12, no. 3, pp. 1–19, Aug. 2013, doi: 10.2136/vzj2013.05.0089.
[61] A. Al-Yaari et al., "Testing regression equations to derive long-term global soil moisture datasets from passive microwave observations," *Remote Sens. Environ.*, vol. 180, pp. 453–464, Jul. 2016, doi: 10.1016/j.rse.2015.11.022.
[62] C. Massari et al., "Near real time de-noising of satellite-based soil moisture retrievals: An intercomparison among three different techniques," *Remote Sens. Environ.*, vol. 198, pp. 17–29, Sep. 2017, doi: 10.1016/j.rse.2017.05.037.
[63] P. M. Atkinson, C. Jeganathan, J. Dash, and C. Atzberger, "Inter-comparison of four models for smoothing satellite sensor time-series data to estimate vegetation phenology," *Remote Sens. Environ.*, vol. 123, pp. 400–417, Aug. 2012, doi: 10.1016/j.rse.2012.04.001.
[64] P. Song, L. R. Mansaray, J. Huang, and W. Huang, "Mapping paddy Rice agriculture over China using AMSR-E time series data," *ISPRS J. Photogramm. Remote Sens.*, vol. 144, pp. 469–482, Oct. 2018, doi: 10.1016/j.isprsjprs.2018.08.015.
[65] R. H. Reichle et al., "Version 4 of the SMAP Level-4 soil moisture algorithm and data product," *J. Adv. Model. Earth Syst.*, vol. 11, no. 10, pp. 3106–3130, Oct. 2019, doi: 10.1029/2019ms001729.
[66] A. Ducharne, R. D. Koster, M. J. Suarez, M. Stieglitz, and P. Kumar, "A catchment-based approach to modeling land surface processes in a general circulation model: 2. parameter estimation and model demonstration," *J. Geophys. Research: Atmos.*, vol. 105, no. D20, pp. 24823–24838, Oct. 2000, doi: 10.1029/2000jd900328.
[67] Z. Su et al., "The Tibetan Plateau observatory of Plateau scale soil moisture and soil temperature (Tibet-Obs) for quantifying uncertainties in coarse resolution satellite and model products," *Hydrol. Earth Syst. Sci.*, vol. 15, no. 7, pp. 2303–2316, Jul. 2011, doi: 10.5194/hess-15-2303-2011.
[68] Z. Su, P. de Rosnay, J. Wen, L. Wang, and Y. Zeng, "Evaluation of ECMWF's soil moisture analyses using observations on the Tibetan Plateau," *J. Geophys. Res., Atmos.*, vol. 118, no. 11, pp. 5304–5318, Jun. 2013, doi: 10.1002/jgrd.50468.
[69] J. Zheng et al., "Use of a new Tibetan Plateau network for permafrost to characterize satellite-based products errors: An application to soil moisture and freeze/thaw," *Remote Sens. Environ.*, vol. 300, Jan. 2024, Art. no. 113899, doi: 10.1016/j.rse.2023.113899.
[70] R. Panciera et al., "The soil moisture active passive experiments (SMAPEx): Toward soil moisture retrieval from the SMAP mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 490–507, Jan. 2014.
[71] A. Jia, S. Liang, D. Wang, L. Ma, Z. Wang, and S. Xu, "Global hourly, 5 km, all-sky Land Surface Temperature data from 2011 to 2021 based on integrating geostationary and polar-orbiting satellite data," *Earth Syst. Sci. Data*, vol. 15, no. 2, pp. 869–895, Feb. 2023, doi: 10.5194/essd-15-869-2023.
[72] S. Dong, J. Cheng, J. Shi, C. Shi, S. Sun, and W. Liu, "A data fusion method for generating hourly seamless Land Surface Temperature from Himawari-8 AHI data," *Remote Sens.*, vol. 14, no. 20, p. 5170, Oct. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/20/5170
[73] L. Ding et al., "Reconstruction of hourly all-weather Land Surface Temperature by integrating reanalysis data and thermal infrared data from geostationary satellites (RTG)," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5003917.
[74] P. Yu et al., "Global spatiotemporally continuous MODIS Land Surface Temperature dataset," *Scientific Data*, vol. 9, no. 1, p. 143, Apr. 2022, doi: 10.1038/s41597-022-01214-8.
[75] S. C. Freitas, I. F. Trigo, J. Macedo, C. Barroso, R. Silva, and R. Perdigão, "Land Surface Temperature from multiple geostationary satellites," *Int. J. Remote Sens.*, vol. 34, nos. 9–10, pp. 3051–3068, May 2013, doi: 10.1080/01431161.2012.716925.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:59:00 UTC from IEEE Xplore. Restrictions apply.